ENERGY CONVERSION - FINAL EXAM

DC MOTORS AND GENERATORS

20 Questions - Time: 90 minutes

1. A DC shunt motor is operating at steady state under rated conditions. If the load torque suddenly DOUBLES while maintaining constant armature and field voltages, what immediately happens to the armature current and motor speed?

   A) Armature current doubles, speed remains approximately constant.
   B) Armature current increases significantly, speed decreases slightly.
   C) Armature current doubles, speed halves.
   D) Armature current remains constant, speed decreases.
   E) Armature current increases slightly, speed drops significantly.

2. Among the following factors, which has the MOST SIGNIFICANT effect on the no-load speed of a DC shunt motor at constant terminal voltage?

   A) Armature resistance
   B) Field circuit resistance
   C) Brush contact resistance
   D) Mechanical friction losses
   E) Core loss in the armature

3. A DC series motor is connected to a constant voltage source and runs satisfactorily under load. However, when the mechanical load is completely removed, the motor speed increases rapidly and dangerously. The MOST PROBABLE cause of this behavior is:

   A) Excessive armature reaction
   B) Open field winding circuit
   C) The inherent characteristic of series motors
   D) Inadequate interpole design
   E) Saturated main pole faces

4. A separately excited DC generator with compensating windings is operating at rated speed and supplying rated load. If the field current is suddenly increased by 20% while the prime mover maintains constant speed and the load resistance remains unchanged, which of the following describes the new steady-state condition?

   A) Terminal voltage increases by 20%, load current increases by 20%.
   B) Terminal voltage increases by approximately 20%, load current increases by approximately 20%.
   C) Terminal voltage increases by less than 20%, load current increases proportionally.
   D) Terminal voltage increases by 20%, load current remains constant.
   E) Terminal voltage increases by more than 20% due to positive feedback.

5. The PRIMARY purpose of using interpoles (commutating poles) in a DC machine is to:

   A) Increase the flux per pole under heavy load

B) Cancel the voltage induced in coils undergoing commutation
C) Reduce the armature resistance
D) Eliminate flux distortion under the main poles
E) Increase the starting torque

6. A DC shunt generator is operating at its rated terminal voltage with no armature reaction effects. If the speed of the prime mover is reduced by 10% while the field current is kept constant, what approximate change occurs in the terminal voltage at no-load?

A) Decreases by 10%
B) Decreases by more than 10% due to loss of field current
C) Decreases by less than 10% due to saturation effects
D) Remains constant due to voltage regulation
E) Drops to nearly zero because the generator cannot sustain voltage

7. A DC machine with lap winding exhibits unstable operation and circulating currents between parallel armature paths despite proper brush positioning. The problem persists even after checking for proper commutation. The MOST LIKELY cause is:

A) Defective or missing equalizer connections
B) Excessive armature resistance
C) Incorrect interpole polarity
D) Weak residual magnetism
E) Overheated field coils

8. For a DC motor operating under constant load torque, the PRIMARY advantage of armature voltage control over field weakening control for speeds below base speed is:

A) Higher efficiency at low speeds
B) Constant maximum torque capability
C) Better commutation quality
D) Reduced armature current
E) Lower cost of control equipment

9. A cumulatively compounded DC motor is running at steady state. If the series field winding suddenly develops an open circuit while the motor continues to run with constant terminal voltage and load torque, what will happen to the motor speed and armature current?

A) Speed increases significantly, armature current increases.
B) Speed decreases, armature current increases.
C) Speed increases significantly, armature current decreases.
D) Speed remains constant, armature current increases significantly.
E) Motor stalls immediately due to loss of torque.

10. A 6-pole DC motor has a simplex wave winding with 720 conductors. At 1200 rpm, the motor draws 50 A and develops 40 Nm of torque. Neglecting losses, the flux per pole is approximately:

A) 18.5 mWb
B) 22.1 mWb
C) 24.7 mWb
D) 27.5 mWb
E) 33.0 mWb

11. In a DC machine with significant armature reaction but WITHOUT compensating windings, the PRIMARY effect of armature reaction on generator operation under increasing load is:

A) Demagnetizing effect causing reduced generated voltage
B) Cross-magnetizing effect causing neutral plane shift
C) Both demagnetizing and cross-magnetizing effects occurring simultaneously
D) Magnetizing effect increasing the generated voltage
E) No significant effect if interpoles are present

12. A DC shunt generator initially builds up voltage normally to its rated value. However, after running for 30 minutes under constant load, the terminal voltage begins to drop gradually and eventually collapses to a very low value. The field circuit resistance has not changed. The MOST PROBABLE cause is:

A) Overheating of field coils causing increased field resistance above critical value
B) Loss of residual magnetism due to demagnetization
C) Excessive armature reaction at this load level
D) Mechanical failure in the prime mover causing speed reduction
E) Commutator surface contamination causing increased resistance

13. A permanent magnet DC (PMDC) motor is operating at steady state. If the armature voltage is increased by 30% while the mechanical load remains constant, which statement is MOST accurate regarding the new steady-state condition?

A) Speed increases by 30%, torque remains constant, armature current remains constant.
B) Speed increases by approximately 30%, torque remains constant, armature current decreases slightly.
C) Speed increases by more than 30% due to reduced losses.
D) Speed increases by 30%, torque increases, armature current increases.
E) Speed increases by less than 30% due to increased armature reaction.

14. Among all types of DC motors, which exhibits the HIGHEST starting torque for a given starting current?

A) Shunt motor
B) Separately excited motor
C) Series motor
D) Cumulatively compounded motor
E) Permanent magnet motor

15. A 250 V DC shunt motor has RA = 0.2 Ω and RF = 125 Ω. At no-load, the motor runs at 1500 rpm drawing 5 A from the supply. When loaded, the line current increases to 52 A. Neglecting brush drop and assuming constant flux, the approximate loaded speed is:

    A) 1380 rpm
    B) 1426 rpm
    C) 1455 rpm
    D) 1471 rpm
    E) 1500 rpm

16. A DC motor is running at steady state. If the mechanical power output remains constant but the motor efficiency increases, what happens to the electrical power input and the losses?

    A) Input power increases, losses increase
    B) Input power decreases, losses decrease
    C) Input power decreases, losses increase
    D) Input power remains constant, losses decrease
    E) Input power increases, losses remain constant

17. The MOST significant disadvantage of using a differentially compounded DC motor compared to a shunt motor is:

    A) Higher cost due to additional windings
    B) Lower starting torque
    C) Unstable operation with positive feedback between speed and load
    D) Higher maintenance requirements
    E) Excessive sparking at brushes

18. A DC motor's starting current exceeds 15 times its rated current even though a starting resistor is properly connected. All other aspects of the motor appear normal. The MOST LIKELY problem is:

    A) The starting resistor value is too small
    B) The starting resistor is being bypassed or shorted
    C) The field circuit is open
    D) Excessive armature reaction
    E) Mechanical binding in the rotor

19. A DC series motor with negligible armature resistance and saturation effects is running at speed n_1 with current I_1. If the load torque is reduced such that the current becomes $0.5I_1$, the new speed n_2 will be approximately:

    A) $0.5n_1$
    B) $0.71n_1$
    C) n_1
    D) $1.41n_1$
    E) $2n_1$

20. For DC machines operating at high speeds and high power levels, the use of compensating windings provides the MOST significant benefit in:

A) Reducing copper losses in the armature
B) Eliminating the need for interpoles
C) Preventing flux distortion and maintaining constant flux under varying loads
D) Increasing the starting torque
E) Improving speed regulation

FORMULAS PROVIDED

Machine Constants:

$K = ZP/(2\pi a)$
$K' = ZP/(60a)$

Voltage and Torque:

$EA = K\varphi\omega m = K'\varphi nm$
$\tau ind = K\varphi IA$
$Pconv = EAI\ A = \tau ind\omega m$

Circuit Equations:

$VT = EA + IARA$ (motor)
$VT = EA - IARA$ (generator)

Speed Relationships:

$n \propto (VT - IARA)/\varphi$
For series motor: $\varphi \propto IA$, $\tau ind \propto IA^2$

Winding Configurations:

Simplex lap: $a = P$
Simplex wave: $a = 2$

Efficiency:

$\eta = Pout/Pin = (Pin - Losses)/Pin$